

1- Diseño de un controlador para un sistema de segundo orden.

Ec. principales.

$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace.

$$V_e(s) = R I_1(s) + L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{C s}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{C s}$$

Procedimiento algebraico.

$$V_e(s) = R I_1(s) + L s I_1(s) - L s I_2(s) + R I_1(s) - R I_2(s)$$

$$V_e(s) = (L s + 2R) I_1(s) - (L s + R) I_2(s)$$

$$L s I_1(s) - L s I_2(s) + R I_1(s) \rightarrow I_1(s) = \frac{1}{L s + R} \left(L s + \frac{1}{C s} + 3R \right) I_2(s)$$

$$V_s(s) = \left(R + \frac{1}{C s} \right) I_2(s)$$

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* Función de Transferencia.

$$\frac{Vs(s)}{Ve(s)} = \frac{\left(R + \frac{1}{Cs}\right) I_2(s)}{(Ls+2R) \left[\frac{1}{Ls+R} \left(\frac{1}{Cs} \right) I_2(s) \right] - (Ls+R) I_2(s)}$$

$$\frac{Vs(s)}{Ve(s)} = \frac{\left(\frac{RCs+1}{Cs} \right) I_2(s)}{(Ls+2R) \left[\frac{1}{Ls+R} \left(\frac{LCs^2+3RCs+1}{Cs} \right) I_2(s) \right] - (Ls+R) I_2(s)}$$

$$\frac{Vs(s)}{Ve(s)} = \frac{\left(\frac{RCs+1}{Cs} \right) I_2(s)}{(Ls+2R) \left[\frac{LCs^2+3RCs+1}{CLs^2+RCs} \right] I_2(s) - (Ls+R) I_2(s)}$$

$$\frac{Vs(s)}{Ve(s)} = \frac{\left(\frac{RCs+1}{Cs} \right) I_2(s)}{\left[\frac{L^2Cs^3+3RLCs^2+Ls}{CLs^2+RCs} - \frac{2LCRs^2+6R^2Cs+2R}{CLs^2+RCs} - (Ls+R) \right] I_2(s)}$$

$$\frac{Vs(s)}{Ve(s)} = \frac{\left(\frac{RCs+1}{Cs} \right) I_2(s)}{\left[\frac{L^2Cs^3+3RLCs^2+Ls+2LCRs^2+6R^2Cs+2R-CL^2s^3-RLCs^2-CLs^2-R^2Cs}{CLs^2+RCs} \right] I_2(s)}$$

$$\frac{Vs(s)}{Ve(s)} = \frac{\left(\frac{RCs+1}{Cs} \right) I_2(s)}{\left[\frac{3RLCs^2+(5R^2C+L)s+2R}{CLs^2+RCs} \right] I_2(s)}$$

$$\frac{Vs(s)}{Ve(s)} = \frac{CLRs^2+(CR^2+L)s+R}{3CLRs^2+(5R^2C+L)s+2R}$$

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* Error estacionario.

$$e(s) = \lim_{s \rightarrow 0} s V_e(s) \left(1 - \frac{V(s)}{V_e(s)} \right)$$

$$e(s) = \lim_{s \rightarrow 0} s \frac{1}{s} \left[1 - \frac{CLRs^2 + (CR^2 + L)s + R}{3CLRs^2 + (5CR^2 + L)s + 2R} \right]$$

$$e(s) = \lim_{s \rightarrow 0} \left[1 - \frac{R}{2R} \right] = \lim_{s \rightarrow 0} \left[1 - \frac{1}{2} \right]$$

$$e(s) = 0.5$$

$$e(t) = 0.5 \text{ V}$$

* Estabilidad en lazo abierto.

$$R = 15 \text{ K}\Omega ; C = 330 \mu\text{F} \text{ y } L = 680 \mu\text{H}$$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 3CLR = 0.010098$$

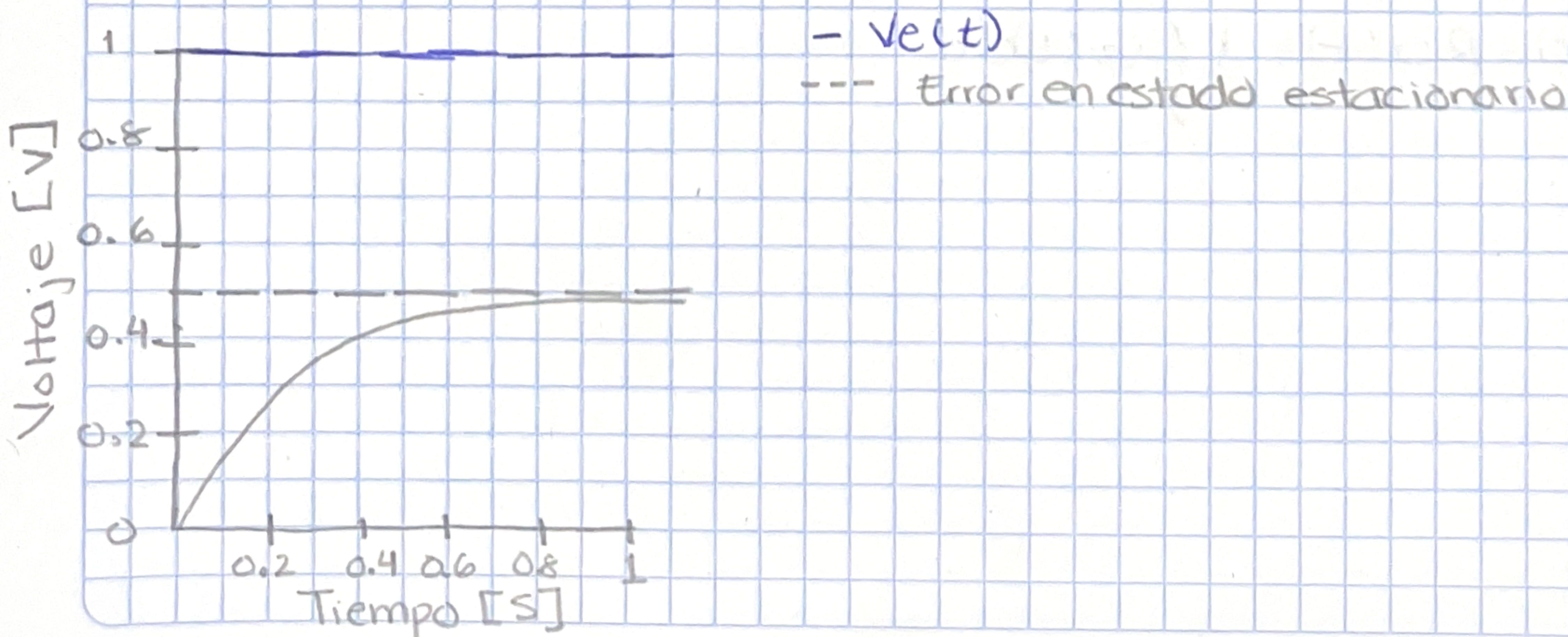
$$b = 5R^2C + L = 371,250.0007$$

$$c = 2R = 30000$$

$$\lambda_1 = -80.80 \text{ m} \text{ y } \lambda_2 = -36.76 \text{ M}$$

Por lo tanto, el sistema tiene una respuesta sobreamortiguada y estable.

* Bosquejo.



* Modelo de ecuaciones integro-diferenciales.

El modelo se expresa despejando las variables dependientes (corrientes)

$$V_e(t) = 2Ri_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - Ri_2(t)$$

$$i_1(t) = \left[V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) = Ri_2(t) + Ri_2(t) + Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$i_1(t) = \left[V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$i_1(t) = \left[V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{2R}$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$